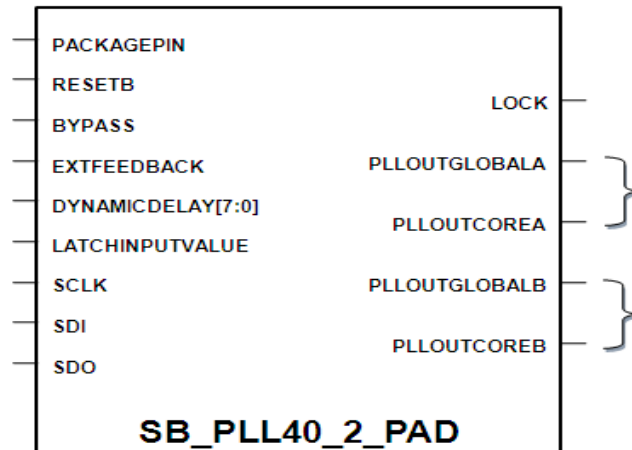


DIVF	Feedback divider	0,1,...,63	control the output frequency, depending on the FEEDBACK_PATH setting.
DIVQ	VCO Divider	1,2,...,6	
FILTER_RANGE	PLL Filter Range	0,1,...,7	
EXTERNAL_DIVIDE_FACTOR	Divide-by factor of a divider in external feedback path	User specified value. Default 1	Specified only when there is a user-implemented divider in the external feedback path.
ENABLE_ICEGATE	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input

SB_PLL40_2_PAD

The SB_PLL40_2_PAD primitive should be used when the source clock of the PLL is driven by an input pad that is located in the bottom IO bank (IO Bank 2) or the top IO bank (IO Bank 0), and in addition to the PLL output, the source clock is also required inside the FPGA.



Ports

PACKAGEPIN: PLL REFERENCE CLOCK pin that serves as the input to the SB_PLL40_2_PAD primitive.

PLLOUTGLOBALA: The signal on PACKAGEPIN appears on the FPGA at this pin, and drives a global clock network on the FPGA. Do not use this pin in an external feedback path to the PLL.

PLLOUTCOREA: The signal on PACKAGEPIN appears on the FPGA at this pin, which drives regular FPGA routing. Do not use this pin in an external feedback path to the PLL.

PLLOUTGLOBALB: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCOREB: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBAL port.

LOCK: Output port, when HIGH, indicates that the signal on PLLOUTGLOBALB/PLLOUTCOREB is locked to the PLL source on PACKAGEPIN.

EXTFEEDBACK: External feedback input to PLL. Enabled when the FEEDBACK_PATH parameter is set to EXTERNAL.

DYNAMICDELAY: 4 bit input bus that enables dynamic control of the delay contributed by the Fine Delay Adjust Block. The Fine Delay Adjust Block is used when there is a need to adjust the phase alignment of PLLOUTGLOBAL/PLLOUTCORE with respect to PACKAGEPIN. The DYNAMICDELAY port controls are enabled when the DELAY_ADJUSTMENT_MODE parameter is set to DYNAMIC.

RESET: Active low input that asynchronously resets the PLL.

BYPASS: Input signal, when asserted, connects the signal on PACKAGEPIN to PLLOUTCORE/PLLOUTGLOBAL pins.

LATCHINPUTVALUE: Active high input, when enabled, forces the PLL into low-power mode. The PLLOUTGLOBALA/PLLOUTCOREA pins are held static at their last value only when the parameter ENABLE_ICEGATE_PORTA is set to '1', and the LATCHINPUTVALUE signal is asserted. The PLLOUTGLOBALB/PLLOUTCOREB pins are held static at their last value only when the parameter ENABLE_ICEGATE_PORTB is set to '1', and the LATCHINPUTVALUE signal is asserted.

SCLK, SDI, SDO: These pins are used only for internal testing purposes, and need not be instantiated by users.

Parameters

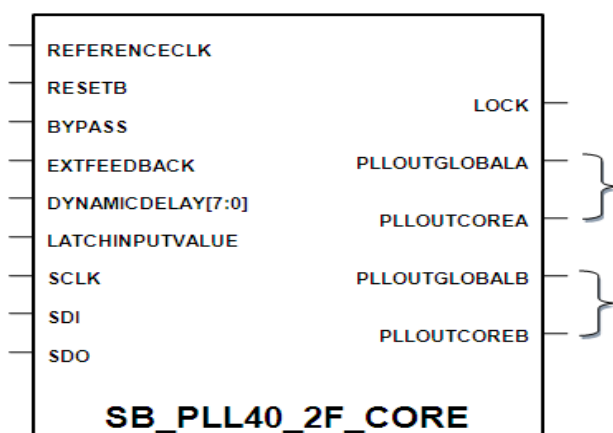
The SB_PLL40_2_PAD primitive requires configuration through the specification of the following parameters. It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

Parameter Name	Description	Parameter Value	Description
FEEDBACK_PATH	Selects the feedback path to the PLL	SIMPLE	Feedback is internal to the PLL, directly from VCO
		DELAY	Feedback is internal to the PLL, through the Fine Delay Adjust Block
		PHASE_AND_DELAY	Feedback is internal to the PLL, through the Phase Shifter and the Fine Delay Adjust Block
		EXTERNAL	Feedback path is external to the PLL, and connects to EXTFEEDBACK pin. Also uses the Fine Delay Adjust Block.
DELAY_ADJUSTMENT_MODE_FEEDBACK	Selects the mode for the Fine Delay Adjust block in the feedback path	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_FEEDBACK parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[3:0] pins
FDA_FEEDBACK	Sets a constant value for the Fine Delay Adjust Block in the feedback path	0, 1,...,15	The PLLOUTGLOBAL & PLLOUTCORE signals are delay compensated by $(n+1)*150$ ps, where $n = \text{FDA_FEEDBACK}$ only if the setting of the DELAY_ADJUSTMENT_MODE_FEEDBACK is FIXED.
DELAY_ADJUSTMENT_MODE_RELATIVE	Selects the mode for the Fine Delay Adjust block	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_RELATIVE parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[7:4] pins
FDA_RELATIVE	Sets a constant value for the Fine Delay Adjust Block	0, 1,...,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are delayed w.r.t. the Port B signals, by $(n+1)*150$ ps, where $n = \text{FDA_RELATIVE}$. Used if DELAY_ADJUSTMENT_MODE_RELATIVE is "FIXED".
SHIFTREG_DIV_MODE	Selects shift register configuration	0,1	Used when FEEDBACK_PATH is "PHASE_AND_DELAY". 0→Divide by 4 1→Divide by 7
PLLOUT_SELECT_PORTB	Selects the signal to be output at the PLLOUTCOREB and PLLOUTGLOBALB ports	SHIFTREG_0deg	0° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output to PortB. No phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output to PORTB. No phase shift.

DIVR	REFERENCECLK divider	0,1,2,...,15	These parameters are used to control the output frequency, depending on the FEEDBACK_PATH setting.
DIVF	Feedback divider	0,1,...,63	
DIVQ	VCO Divider	1,2,...,6	
FILTER_RANGE	PLL Filter Range	0,1,...,7	
EXTERNAL_DIVIDE_FACTOR	Divide-by factor of a divider in external feedback path	User specified value. Default 1	Specified only when there is a user-implemented divider in the external feedback path.
ENABLE_ICEGATE_PORTA	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input
ENABLE_ICEGATE_PORTB	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input

SB_PLL40_2F_CORE

The SB_PLL40_2F_CORE primitive should be used when PLL is used to generate 2 different output frequencies, and the source clock of the PLL is driven by FPGA routing i.e. when the PLL source clock originates on the FPGA.



Ports

REFERENCECLK: PLL source clock that serves as the input to the SB_PLL40_2F_CORE primitive.

PLLOUTGLOBALA: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCOREA: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBALA port.

PLLOUTGLOBALB: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCOREB: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBALB port.

LOCK: Output port, when HIGH, indicates that the signal on PLLOUTGLOBALB/PLLOUTCOREB is locked to the PLL source on PACKAGEPIN.

EXTFEEDBACK: External feedback input to PLL. Enabled when the *FEEDBACK_PATH* parameter is set to *EXTERNAL*.

DYNAMICDELAY: 4 bit input bus that enables dynamic control of the delay contributed by the Fine Delay Adjust Block. The Fine Delay Adjust Block is used when there is a need to adjust the phase alignment of *PLLOUTGLOBAL/PLLOUTCORE* with respect to *REFERENCECLK*. The *DYNAMICDELAY* port controls are enabled when the *DELAY_ADJUSTMENT_MODE* parameter is set to *DYNAMIC*.

RESETB: Active low input that asynchronously resets the PLL.

BYPASS: Input signal, when asserted, connects the signal on *REFERENCECLK* to *PLLOUTCORE/PLLOUTGLOBAL* pins.

LATCHINPUTVALUE: Active high input, when enabled, forces the PLL into low-power mode. The *PLLOUTGLOBALA/PLLOUTCOREA* pins are held static at their last value only when the parameter *ENABLE_ICEGATE_PORTA* is set to '1', and the *LATCHINPUTVALUE* signal is asserted. The *PLLOUTGLOBALB/PLLOUTCOREB* pins are held static at their last value only when the parameter *ENABLE_ICEGATE_PORTB* is set to '1', and the *LATCHINPUTVALUE* signal is asserted.

SCLK, *SDI*, *SDO*: These pins are used only for internal testing purposes, and need not be instantiated by users.

Parameters

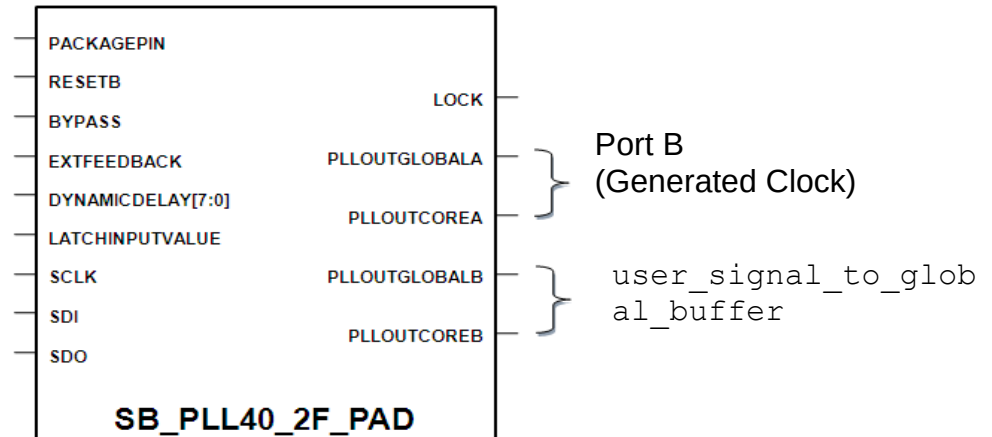
The *SB_PLL40_2F_CORE* primitive requires configuration through the specification of the following parameters. It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.

Parameter Name	Description	Parameter Value	Description
FEEDBACK_PATH	Selects the feedback path to the PLL	SIMPLE	Feedback is internal to the PLL, directly from VCO
		DELAY	Feedback is internal to the PLL, through the Fine Delay Adjust Block
		PHASE_AND_DELAY	Feedback is internal to the PLL, through the Phase Shifter and the Fine Delay Adjust Block
		EXTERNAL	Feedback path is external to the PLL, and connects to EXTFEEDBACK pin. Also uses the Fine Delay Adjust Block.
DELAY_ADJUSTMENT_MODE_FEEDBACK	Selects the mode for the Fine Delay Adjust block in the feedback path	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_FEEDBACK parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[3:0] pins
FDA_FEEDBACK	Sets a constant value for the Fine Delay Adjust Block in the feedback path	0, 1,...,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are delay compensated by (n+1)*150 ps, where n = FDA_FEEDBACK only if the setting of the DELAY_ADJUSTMENT_MODE_FEEDBACK is FIXED.
DELAY_ADJUSTMENT_MODE_RELATIVE	Selects the mode for the Fine Delay Adjust block	FIXED	Delay of the Fine Delay Adjust Block is fixed, the value is specified by the FDA_RELATIVE parameter setting
		DYNAMIC	Delay of Fine Delay Adjust Block is determined by the signal value at the DYNAMICDELAY[7:4] pins
FDA_RELATIVE	Sets a constant value for the Fine Delay Adjust Block	0, 1,...,15	The PLLOUTGLOBALA & PLLOUTCOREA signals are delayed w.r.t. the Port B signals, by (n+1)*150 ps, where n = FDA_RELATIVE. Used if DELAY_ADJUSTMENT_MODE_RELATIVE is "FIXED".
SHIFTREG_DIV_MODE	Selects shift register configuration	0,1	Used when FEEDBACK_PATH is "PHASE_AND_DELAY". 0→Divide by 4 1→Divide by 7
PLLOUT_SELECT_PORTA	Selects the signal to be output at the PLLOUTCOREA and PLLOUTGLOBALA ports	SHIFTREG_0deg	0° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output to PortA. No phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output to PORTA. No phase shift.

PLLOUT_SELECT_PORTB	Selects the signal to be output at the PLLOUTCOREB and PLLOUTGLOBALB ports	SHIFTREG_0deg	0° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY"
		SHIFTREG_90deg	90° phase shift only if the setting of FEEDBACK_PATH is "PHASE_AND_DELAY" and SHIFTREG_DIV_MODE=0
		GENCLK	The internally generated PLL frequency will be output to PortB. No phase shift.
		GENCLK_HALF	The internally generated PLL frequency will be divided by 2 and then output to PORTB. No phase shift.
DIVR	REFERENCECLK divider	0,1,2,...,15	These parameters are used to control the output frequency, depending on the FEEDBACK_PATH setting.
DIVF	Feedback divider	0,1,...,63	
DIVQ	VCO Divider	1,2,...,6	
FILTER_RANGE	PLL Filter Range	0,1,...,7	
EXTERNAL_DIVIDE_FACTOR	Divide-by factor of a divider in external feedback path	User specified value. Default 1	Specified only when there is a user-implemented divider in the external feedback path.
ENABLE_ICEGATE_PORTA	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input
ENABLE_ICEGATE_PORTB	Enables the PLL power-down control	0	Power-down control disabled
		1	Power-down controlled by LATCHINPUTVALUE input

SB_PLL40_2F_PAD

The SB_PLL40_2F_PAD primitive should be used when the PLL is used to generate 2 different output frequencies, and the source clock of the PLL is driven by an input pad located in the bottom IO bank (IO Bank 2) or the top IO bank (IO Bank 0).



Ports

PACKAGEPIN: PLL REFERENCE CLOCK pin that serves as the input to the SB_PLL40_2F_PAD primitive.

PLLOUTGLOBALA: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCOREA: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBALA port.

PLLOUTGLOBALB: Output clock generated by the PLL, drives a global clock network on the FPGA.

PLLOUTCOREB: Output clock generated by the PLL, drives regular FPGA routing. The frequency generated on this output is the same as the frequency of the clock signal generated on the PLLOUTGLOBALB port.

LOCK: Output port, when HIGH, indicates that the signal on PLLOUTGLOBALB/PLLOUTCOREB is locked to the PLL source on PACKAGEPIN.

EXTFEEDBACK: External feedback input to PLL. Enabled when the FEEDBACK_PATH parameter is set to EXTERNAL.

DYNAMICDELAY: 4 bit input bus that enables dynamic control of the delay contributed by the Fine Delay Adjust Block. The Fine Delay Adjust Block is used when there is a need to adjust the phase alignment of PLLOUTGLOBAL/PLLOUTCORE with respect to PACKAGEPIN. The DYNAMICDELAY port controls are enabled when the DELAY_ADJUSTMENT_MODE parameter is set to DYNAMIC.

RESETB: Active low input that asynchronously resets the PLL.

BYPASS: Input signal, when asserted, connects the signal on PACKAGEPIN to PLLOUTCORE/PLLOUTGLOBAL pins.

LATCHINPUTVALUE: Active high input, when enabled, forces the PLL into low-power mode. The PLLOUTGLOBALA/PLLOUTCOREA pins are held static at their last value only when the parameter ENABLE_ICEGATE_PORTA is set to '1', and the LATCHINPUTVALUE signal is asserted. The PLLOUTGLOBALB/PLLOUTCOREB pins are held static at their last value only when the parameter ENABLE_ICEGATE_PORTB is set to '1', and the LATCHINPUTVALUE signal is asserted.

SCLK, SDI, SDO: These pins are used only for internal testing purposes, and need not be instantiated by users.

Parameters

The SB_PLL40_2F_PAD primitive requires configuration through the specification of the following parameters. It is strongly recommended that the configuration of the PLL primitives be accomplished through the use of the PLL Configuration tool that is offered as part of the iCEcube2 software.